Volume disaster recovery preparation workflow

System Manager Classic

NetApp
February 11, 2024

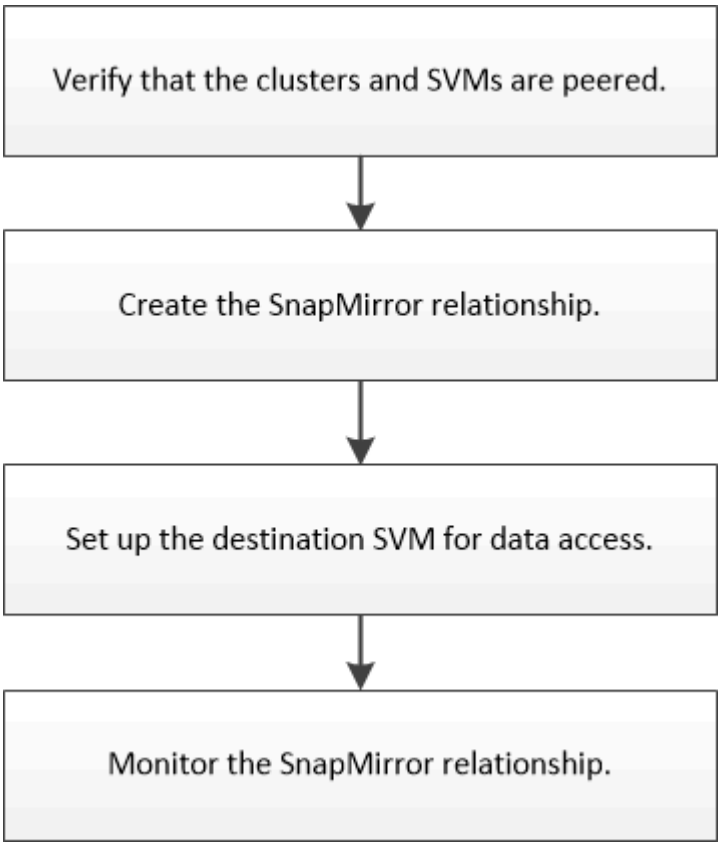
This PDF was generated from https://docs.netapp.com/us-en/ontap-sm-classic/volume-disaster-prep/task_verifying_cluster_peer_relationship.html on February 11, 2024. Always check docs.netapp.com for the latest.

Table of Contents

Volume disaster recovery preparation workflow . . . 1
Verify the cluster peer relationship and SVM peer relationship . . . 1
Create the SnapMirror relationship (Beginning with ONTAP 9.3) . . . 2
Create the SnapMirror relationship (ONTAP 9.2 or earlier) . . . 4
Set up the destination SVM for data access . . . 7
Monitor the status of SnapMirror data transfers . . . 8

Volume disaster recovery preparation workflow

Preparing volumes for disaster recovery involves verifying the cluster peer relationship, creating the SnapMirror relationship between volumes residing on peered clusters, setting up the destination SVM for data access, and monitoring the SnapMirror relationship periodically.



Additional documentation is available to help you activate the destination volume to test the disaster recovery setup or when a disaster occurs. You can also learn more about how to reactivate the source volume after the disaster.

Volume disaster recovery

+ Describes how to quickly activate a destination volume after a disaster and then reactivate the source volume in ONTAP.

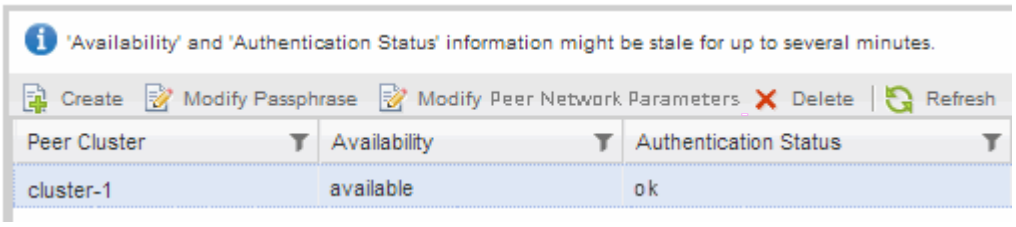
Verify the cluster peer relationship and SVM peer relationship

Before you set up a volume for disaster recovery, you must verify that the source and destination clusters are peered and are communicating with each other through the peer relationship.

Procedure

- If you are running ONTAP 9.3 or later, perform the following steps to verify the cluster peer relationship and SVM peer relationship:
  a. Click **Configuration** > **Cluster Peers**.
  b. Verify that the peered cluster is authenticated and is available.

  c. Click **Configuration** > **SVM Peers**.
  d. Verify that the destination SVM is peered with the source SVM.
- If you are running ONTAP 9.2 or earlier, perform the following steps to verify the cluster peer relationship and SVM peer relationship:
  a. Click the **Configurations** tab.
  b. In the **Cluster Details** pane, click **Cluster Peers**.
  c. Verify that the peered cluster is authenticated and available.



  d. Click the **SVMs** tab and select the source SVM.
  e. In the **Peer Storage Virtual Machines** area, verify the destination SVM is peered with the source SVM.

     If you do not see any peered SVM in this area, you can create the SVM peer relationship when creating the SnapMirror relationship.

Creating the SnapMirror relationship (ONTAP 9.2 or earlier)

Create the SnapMirror relationship (Beginning with ONTAP 9.3)

You must create a SnapMirror relationship between the source volume on one cluster and the destination volume on the peered cluster for replicating data for disaster recovery.

Before you begin

- The destination aggregate must have available space.
- Both the clusters must be configured and set up appropriately to meet the requirements of your environment for user access, authentication, and client access.

About this task

You must perform this task from the **source** cluster.

Steps

1. Click **Storage** > **Volumes**.
2. Select the volume for which you want to create a mirror relationship, and then click **Actions** > **Protect**.
3. In the **Relationship Type** section, select **Mirror** from the **Relationship Type** drop-down list.
4. In the **Volumes: Protect Volumes** page, provide the following information:
   a. Select **Mirror** as the relationship type.
   b. Select the destination cluster, destination SVM, and the suffix for the name of the destination volume.

      Only peered SVMs and allowed SVMs are listed under destination SVMs.

   c. Click .
   d. In the **Advanced Options** dialog box, verify that `MirrorAllSnapshots` is set as the protection policy.

      `DPDefault` and `MirrorLatest` are the other default protection policies that are available for SnapMirror relationships.

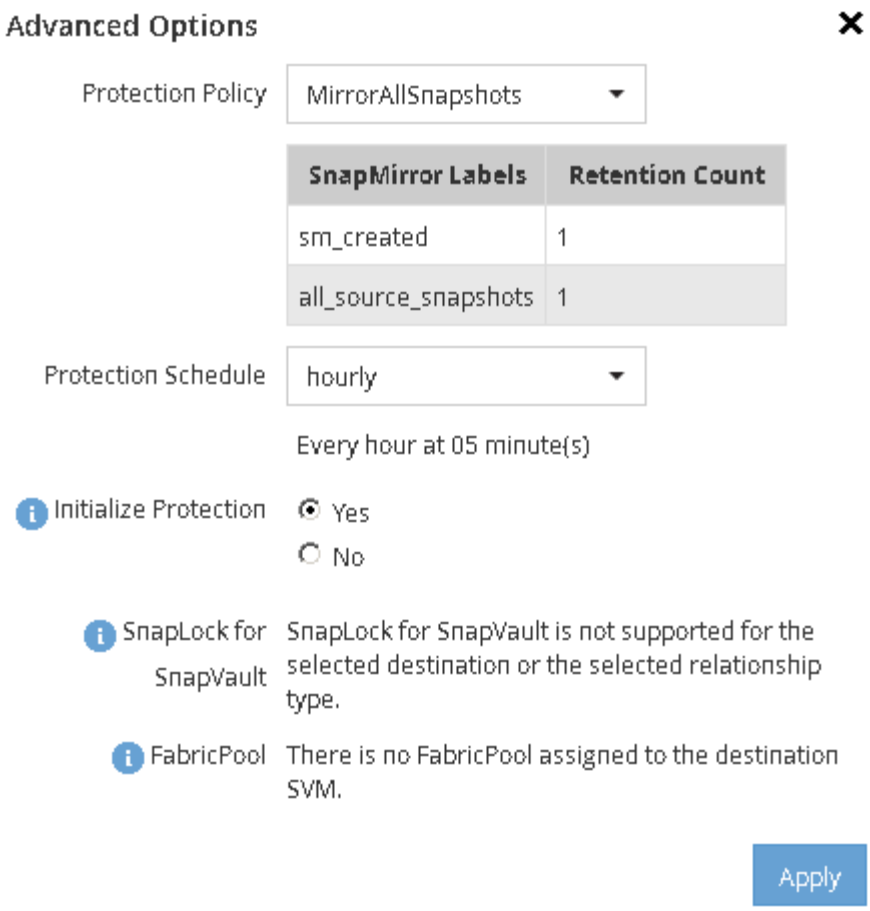
   e. Select a protection schedule.

      By default, the `hourly` schedule is selected.

   f. Verify that **Yes** is selected for initializing the SnapVault relationship.

      All of the data protection relationships are initialized by default. Initializing the SnapMirror relationship ensures that the destination volume has a baseline to start protecting the source volume.

   g. Click **Apply** to save the changes.



5. Click **Save** to create the SnapMirror relationship.
6. Verify that the relationship status of the SnapMirror relationship is in the `Snapmirrored` state.

   a. Navigate to the **Volumes** window, and then select the volume that the volume for which you created the SnapMirror relationship.

   b. Double-click the volume to view the volume details, and then click **PROTECTION** to view the data protection status of the volume.

What to do next

You must make a note of the settings for the source volume such as thin provisioning, deduplication, compression, and autogrow. You can use this information to verify the destination volume settings when you break the SnapMirror relationship.

Create the SnapMirror relationship (ONTAP 9.2 or earlier)

You must create a SnapMirror relationship between the source volume on one cluster and the destination volume on the peered cluster for replicating data for disaster recovery.

Before you begin

- You must have the cluster administrator user name and password for the destination cluster.
- The destination aggregate must have available space.
- Both the clusters must be configured and set up appropriately to meet the requirements of your environment for user access, authentication, and client access.

About this task

You must perform this task from the **source** cluster.

Steps

1. Click **Storage** > **SVMs**.
2. Select the SVM, and then click **SVM Settings**.
3. Click the **Volumes** tab.
4. Select the volume for which you want to create a mirror relationship, and then click **Protect**.

   The Create Protection Relationship window is displayed.

5. In the **Relationship Type** section, select **Mirror** from the **Relationship Type** drop-down list.
6. In the **Destination Volume** section, select the peered cluster.
7. Specify the SVM for the destination volume:

If the SVM is…	Then…
Peered	Select the peered SVM from the list.
Not peered	a. Select the SVM. b. Click **Authenticate**. c. Enter the cluster administrator's credentials of the peered cluster, and then click **Create**.

8. Create a new destination volume:

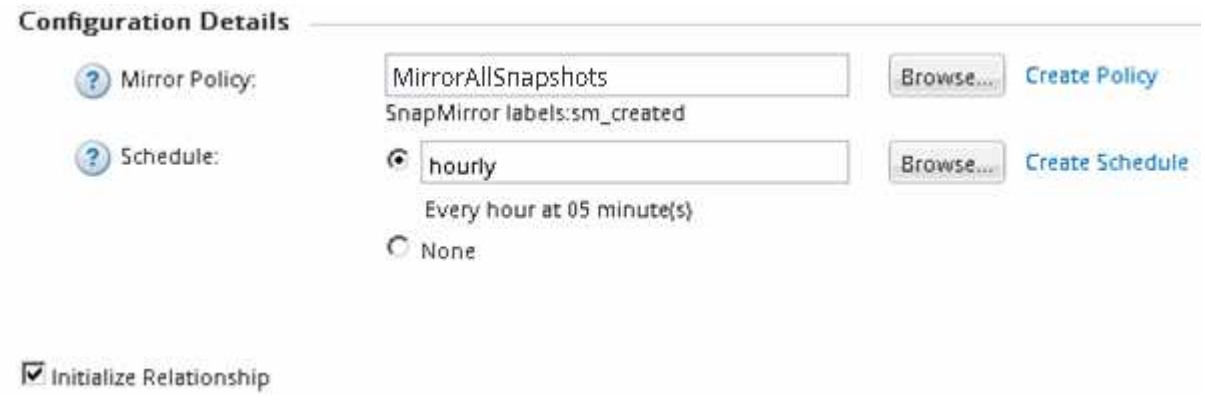
   a. Select the **New Volume** option.
   b. Use the default volume name or specify a new volume name.
   c. Select the destination aggregate.

9. In the **Configuration Details** section, select **MirrorAllSnapshots** as the mirror policy.

`DPDefault` and `MirrorLatest` are the other default mirror policies that are available for SnapMirror relationships.

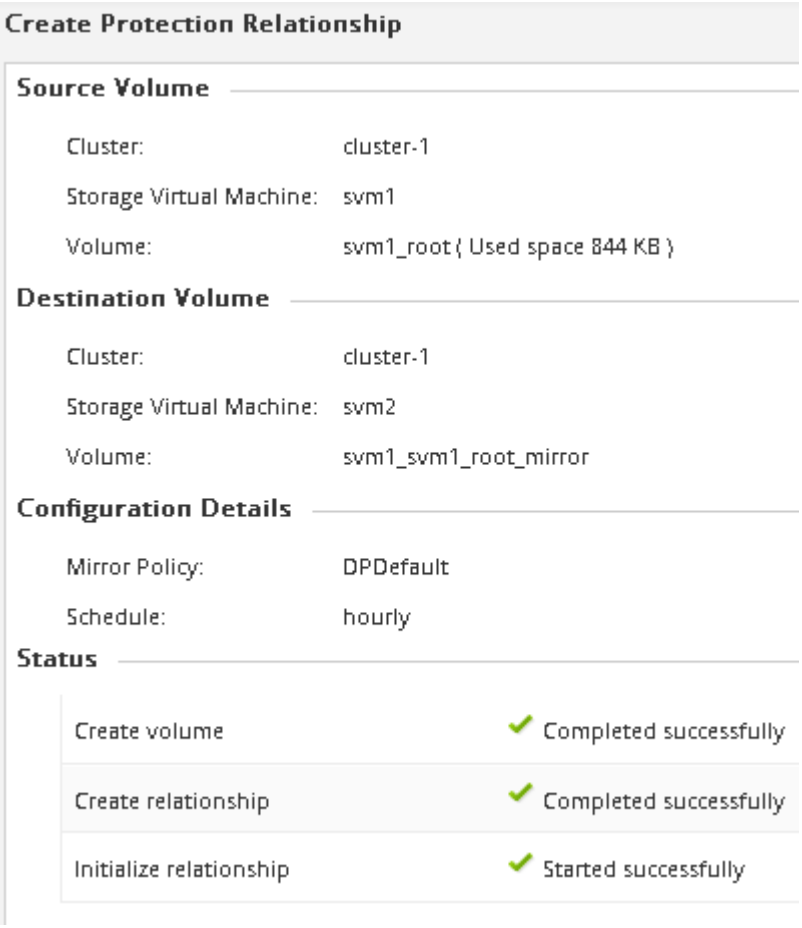
10. Select a protection schedule from the list of schedules.
11. Ensure that the **Initialize Relationship** check box is selected, and then click **Create**.

    Initializing the SnapMirror relationship ensures that the destination volume has a baseline to start protecting the source volume.



    The relationship is initialized by starting a baseline transfer of data from the source volume to the destination volume.

    The initialization operation might take some time. The Status section shows the status of each job.



12. Verify the relationship status of the SnapMirror relationship:
    a. Select the volume for which you created the SnapMirror relationship from the **Volumes** list, and then click **Data Protection**.
    b. In the **Data Protection** tab, verify that the SnapMirror relationship that you created is listed and that the relationship state is `Snapmirrored`.

What to do next

You must make a note of the settings for the source volume such as thin provisioning, deduplication, compression, and autogrow. You can use this information to verify the destination volume settings when you break the SnapMirror relationship.

Set up the destination SVM for data access

You can minimize data access disruption when activating the destination volume by setting up required configurations such as LIFs, CIFS shares, and export policies for the NAS environment, and LIFs and initiator groups for the SAN environment on the SVM containing the destination volume.

About this task

You must perform this task on the **destination** cluster for the SVM containing the destination volume.

Procedure

- NAS environment:
    a. Create NAS LIFs.
    b. Create CIFS shares with the same share names that were used on the source.
    c. Create appropriate NFS export policies.
    d. Create appropriate quota rules.
- SAN environment:
    a. Create SAN LIFs.
    b. **Optional:** Configure portsets.
    c. Configure initiator groups.
    d. For FC, zone the FC switches to enable the SAN clients to access the LIFs.

What to do next

If any changes were made on the SVM containing the source volume, you must replicate the changes manually on the SVM containing the destination volume.

Related information

ONTAP 9 Documentation Center

Monitor the status of SnapMirror data transfers

You should periodically monitor the status of the SnapMirror relationships to ensure that the SnapMirror data transfers are occurring as per the specified schedule.

About this task

You must perform this task from the **destination** cluster.

Steps

1. Depending on the System Manager version that you are running, perform one of the following steps:
   - ONTAP 9.4 or earlier: Click **Protection** > **Relationships**.
   - Beginning with ONTAP 9.5: Click **Protection** > **Volume Relationships**.
2. Select the SnapMirror relationship between the source and the destination volumes, and then verify the status in the **Details** bottom tab.

   The Details tab displays the health status of the SnapMirror relationship and shows the transfer errors and lag time.

   - The Is Healthy field must display `Yes`.

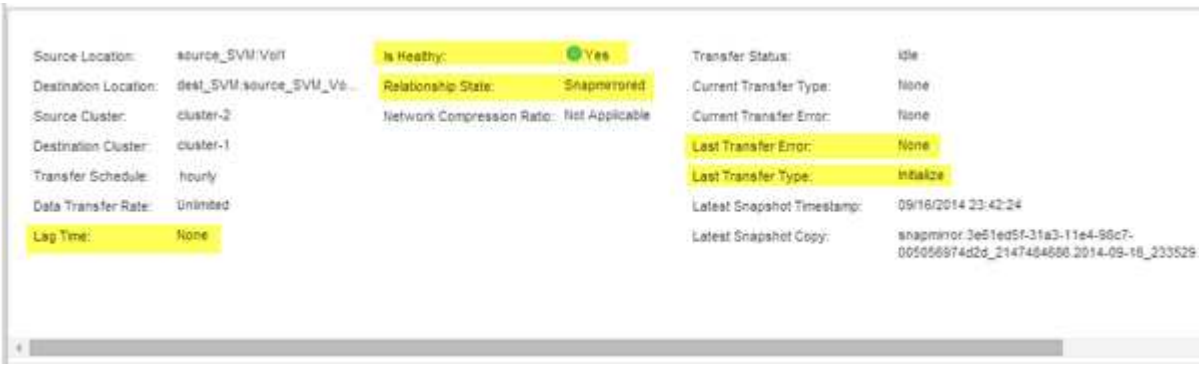
     For most SnapMirror data transfer failures, the field displays `No`. In some failure cases, however, the field continues to display `Yes`. You must check the transfer errors in the Details section to ensure that no data transfer failure occurred.

   - The Relationship State field must display `Snapmirrored`.
   - The Lag Time must be no more than the transfer schedule interval.

     For example, if the transfer schedule is hourly, then the lag time must not be more than an hour.

     You should troubleshoot any issues in the SnapMirror relationships.

     NetApp Technical Report 4015: SnapMirror Configuration and Best Practices for ONTAP 9.1, 9.2



Copyright information

Copyright © 2024 NetApp, Inc. All Rights Reserved. Printed in the U.S. No part of this document covered by copyright may be reproduced in any form or by any means—graphic, electronic, or mechanical, including photocopying, recording, taping, or storage in an electronic retrieval system—without prior written permission of the copyright owner.

Software derived from copyrighted NetApp material is subject to the following license and disclaimer:

THIS SOFTWARE IS PROVIDED BY NETAPP “AS IS” AND WITHOUT ANY EXPRESS OR IMPLIED WARRANTIES, INCLUDING, BUT NOT LIMITED TO, THE IMPLIED WARRANTIES OF MERCHANTABILITY AND FITNESS FOR A PARTICULAR PURPOSE, WHICH ARE HEREBY DISCLAIMED. IN NO EVENT SHALL NETAPP BE LIABLE FOR ANY DIRECT, INDIRECT, INCIDENTAL, SPECIAL, EXEMPLARY, OR CONSEQUENTIAL DAMAGES (INCLUDING, BUT NOT LIMITED TO, PROCUREMENT OF SUBSTITUTE GOODS OR SERVICES; LOSS OF USE, DATA, OR PROFITS; OR BUSINESS INTERRUPTION) HOWEVER CAUSED AND ON ANY THEORY OF LIABILITY, WHETHER IN CONTRACT, STRICT LIABILITY, OR TORT (INCLUDING NEGLIGENCE OR OTHERWISE) ARISING IN ANY WAY OUT OF THE USE OF THIS SOFTWARE, EVEN IF ADVISED OF THE POSSIBILITY OF SUCH DAMAGE.

NetApp reserves the right to change any products described herein at any time, and without notice. NetApp assumes no responsibility or liability arising from the use of products described herein, except as expressly agreed to in writing by NetApp. The use or purchase of this product does not convey a license under any patent rights, trademark rights, or any other intellectual property rights of NetApp.

The product described in this manual may be protected by one or more U.S. patents, foreign patents, or pending applications.

LIMITED RIGHTS LEGEND: Use, duplication, or disclosure by the government is subject to restrictions as set forth in subparagraph (b)(3) of the Rights in Technical Data -Noncommercial Items at DFARS 252.227-7013 (FEB 2014) and FAR 52.227-19 (DEC 2007).

Data contained herein pertains to a commercial product and/or commercial service (as defined in FAR 2.101) and is proprietary to NetApp, Inc. All NetApp technical data and computer software provided under this Agreement is commercial in nature and developed solely at private expense. The U.S. Government has a non-exclusive, non-transferrable, nonsublicensable, worldwide, limited irrevocable license to use the Data only in connection with and in support of the U.S. Government contract under which the Data was delivered. Except as provided herein, the Data may not be used, disclosed, reproduced, modified, performed, or displayed without the prior written approval of NetApp, Inc. United States Government license rights for the Department of Defense are limited to those rights identified in DFARS clause 252.227-7015(b) (FEB 2014).

Trademark information

NETAPP, the NETAPP logo, and the marks listed at http://www.netapp.com/TM are trademarks of NetApp, Inc. Other company and product names may be trademarks of their respective owners.